

Matthew Ball

Computer Science

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Education

2024-Current	📌 University of California Irvine, Ph.D. & M.Sc. in Computer Science	3.9/4.0
	<i>Advisor: Prof. Chen Li</i>	
2022-2024	📌 University of California Berkeley, M.Sc. Data Science	3.96/4.0
2019 – 2022	📌 University of La Verne, B.Sc. Computer Science	3.98/4.0
	<i>Achievements: Summa Cum Laude, Honors Degree, Departmental Honors</i>	

Research

2025-present	📌 AI & Data Workflow Benchmark Suite	
	Designing a system-agnostic benchmark for dataflow, training, inference, and RAG pipelines with standardized workloads and metrics.	
	📌 Efficient On-Device LLM Adaptation	
	Tooling for local LLM fine-tuning (SFT, LoRA) and knowledge distillation, cutting cloud-inference cost while preserving privacy.	
2022-2023	📌 Reinforcement Learning for Text-to-SQL and Query Optimization	
	Applied Q-learning and reward shaping to improve text-to-SQL generation and query plan optimization.	
2023-2025	📌 Semantic Search and Vector Retrieval for Mission-Critical IR	
	Benchmarked Microsoft Semantic Kernel and Elasticsearch against production vector search for low-latency retrieval (U.S. Department of Defense).	

Open Source & Projects

2024-present	📌 Apache Texera (incubating) — Core Contributor	apache/texera
	162+ merged pull requests to a distributed, browser-based data-science platform, spanning backend performance (Postgres pooling), full-stack UI, and CI tooling; building a columnar Apache Arrow vectorized execution engine. <i>Scala, Python, TypeScript</i> .	
2025-present	📌 Apache Spark — Contributor	apache/spark
	Authored patches to Spark SQL, GraphX, and Spark Connect, including error-condition naming (SPARK-58xxx) and test-port hardening. <i>Scala</i> .	
2026	📌 Python-to-Dataflow Compiler	Texera_Compiler
	Lowers imperative Python ML code into distributed Texera workflows. <i>Python</i> .	

Skills

Languages	📌 Python, C++, Java, TypeScript, JavaScript, SQL, R, LaTeX
AI/ML	📌 PyTorch, Hugging Face, LangChain, scikit-learn, RAG, fine-tuning (SFT/LoRA)
Data & Retrieval	📌 Apache Spark, BigQuery, PostgreSQL/pgvector, Elasticsearch, FAISS, Pinecone
Systems & Cloud	📌 Docker, Kubernetes, Git, Linux, Google Cloud (Vertex AI, GKE), Azure